

# Kumu: Connecting the Philippine global community with live streaming innovation

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Kumu developed  
Philippines' biggest  
participatory social  
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improvements with

## About Kumu

Kumu is the largest social entertainment app in the Philippines, on a mission to champion Filipino voices from around the world by creating a vibrant online culture filled with positivity and possibilities. Launched in August 2018 as a messaging and live stream platform, Kumu has experienced significant growth in its user base over the past four years thanks to creating authentic connections between Filipinos wherever they are in the world. Named after the iconic Tagalog greeting "Kumusta ka?," Kumu embodies the very same warmth and fun-loving

Workspace and Cloud.

spirit of Filipinos, amplified by unique innovations in livestreaming and social media.

Industries: Media & Entertainment

Google Cloud results

- Enables global growth mission with more than two-fold rise in monthly paid users
- Enhances platform stickiness by eliminating weekly outages caused by scaling issues
- Supports stable live streams and user interaction by reducing latency from

15% to in new retent

Location: The Philippines

Kubernetes Engine

Google Cloud (<https://cloud.google.com/>) gine/),  
Google Workspace  
BigQuery (<https://cloud.google.com/bigquery/>)  
(<https://workspace.google.com/products/docs/>)  
Cloud CDN (<https://cloud.google.com/cdn/>)  
Google Drive  
Kubernetes Engine  
(<https://cloud.google.com/kubernetes-engine/>)  
Sheets  
Google Workspace  
(<https://workspace.google.com/>)  
  
Docs  
(<https://workspace.google.com/products/docs/>)  
  
Google Drive  
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Sheets  
(<https://workspace.google.com/products/sheets/>)

250  
milliseconds  
to 100  
milliseconds

- Drives  
DevOps cost  
and  
workload  
efficiencies  
by reducing  
scaling time  
for major  
events from  
15 minutes  
to nearly  
zero

More than 10 million Filipinos are spread around the globe, from the United States to Dubai and Japan, making them one of the largest and most extensive (<https://www.pilotguides.com/study-guides/the-filipino-diaspora/>)

diasporas in the world. In 2018, founders Roland Ros and Remy Dorado launched Kumu (<https://kumu.ph/>) out of a dream to connect Filipinos wherever they are with a participatory social media app that enables them to share their lives and build communities through live streaming videos. The vision was to create an authentically positive space for Filipinos to connect and make a living via respectful live streaming content. Since its inception, Kumu has scaled in the Philippine market, while encompassing its reach to over 55 countries with a strong Filipino diaspora.

And scale Kumu did. Today the app enjoys millions of active monthly users, making it the Philippines' largest social entertainment platform. During the pandemic, as usage soared, Kumu provided a vital emotional lifeline between expat Filipinos and loved-ones back home through an uninterrupted stream of user content.

In order to drive this remarkable global journey, Kumu needed a combination of the most advanced collaboration tools, powerful data analytics, seamless autoscaling power, and minimal latency connectivity, to support both unique product creation and real-time content delivery, whatever the demand.

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*—Ellard Capiral, Vice-  
President of Engineering,  
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By turning to a combination of [Google Workspace](https://workspace.google.com/) (https://workspace.google.com/) productivity tools and [Google Cloud](https://cloud.google.com/) (https://cloud.google.com/) computing, content delivery and data analytics solutions, Kumu unleashed a continuous stream of innovation potential for bringing together the vast and diverse Filipino global community.

"Our dream is to create a worldwide Filipino network through the power of video live streaming," says Ellard Capiral, Vice-President of Engineering, Kumu. "With Google Workspace and Google Cloud, we're able not only to continuously evolve the app to delight users in real time, but also deliver with zero downtime even at the busiest times."

**Winning fans with data-driven  
insight through BigQuery and**

## Sheets synergies

Kumu sees its unique value as being a data-driven firm that captures user needs through smart analytics, then translates this insight into non-stop platform improvements. Through close teamwork among developers scattered across the Philippines, Kumu is constantly seeking to improve user experience by analyzing behavior, such as how often users comment, share videos, or offer gifts on the platform. This enables Kumu to develop content promotion strategies, and new features that cater to specific behavior patterns and trends.

Kumu's Google Cloud journey began a year after its founding in 2018 when it migrated its core workforce to Google Workspace, from a variety of piecemeal productivity solutions. Kumu tapped Google Workspace because of the way it enabled real-time collaboration with no software installations required. This allowed it during the pandemic to rapidly execute a sevenfold increase in its Google Workspace community across its value-chain.

The true transformative power of the migration became clear, says Capiral, when Kumu discovered how BigQuery (<https://cloud.google.com/bigquery>), as its powerful Google Cloud data warehouse, featured a direct export function into Sheets (<https://workspace.google.com/products/sheets/>), its Google Workspace collaborative spreadsheet. Through integration of the two solutions, Kumu gained the ability to create user-friendly formulas and charts to identify trends in BigQuery datasets, for instant actionable insight.

"The symbiosis of powerful tools like BigQuery and Sheets give us precise, user-friendly insight to boost engagement. By understanding how people use the app, we geared our strategy towards the types of content we promote, for significant revenue gain."

*—Ellard Capiral, Vice-  
President of Engineering,  
Kumu*

Building professional user-generated content, we were able to understand what people want better

with BigQuery, which enabled Kumu to rapidly optimize product offerings.

The result has been significant gains in Kumu's customer retention and engagement. Kumu has enjoyed a nearly threefold jump in new user retention rate from 15% to 40%.

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## **Unlimited video storage and world-class privacy protection with Google Workspace**

For a streaming platform with 2 million active monthly users, sharing lives online, it's essential that Kumu enjoys a combination of unlimited video storage capacity, and rock-solid data security for personal identifiable information (PII).

Kumu worked with the Google team to choose the right Google Workspace Enterprise plan to enable both objectives at an affordable price. The startup opted for an Enterprise plan for which it pays a single unlimited storage rate for Drive

(<https://www.google.com/drive/>) across the company, instead of user-by-user, which Capiral says "is cheaper and brings no billing surprises."





*Kumu app*

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The Enterprise plan also gives Kumu advanced security management and compliance controls that protect user data, as it is shared in Google Workspace solutions, such as [Docs](https://workspace.google.com/products/docs/) (<https://workspace.google.com/products/docs/>) and Sheets. In particular, Kumu relies upon a feature that automatically sends an alert to team leaders whenever PII is added to a Google Workspace document. This enables Kumu to edit the document before it is shared with an external partner, or with an unauthorized internal team member.

"We have a critical need to enjoy secure unlimited storage to accommodate a growing global audience, as well as protect the privacy of that audience," says Capiral. "Google Workspace provides us flexible pricing plans to achieve unrivaled capabilities in both, all at an affordable price."

# Planet-scale live streaming capabilities with Google Kubernetes Engine

Kumu was so impressed by the power unleashed by BigQuery and Google Workspace symbiosis that it decided in 2022 to begin a full-scale phased migration of its computing infrastructure to Google Cloud, from a rival global cloud provider.

Kumu was attracted in particular by the prospect of moving to Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE) for unlimited live stream autoscaling, thanks to Google leadership in Kubernetes microservices. The startup was also keen to gain the ultra low-latency content delivery capabilities enabled by Cloud CDN

(<https://cloud.google.com/cdn>), for stable video delivery anywhere in the world.

Even early in the multi-stage migration, Capiral says, "the advantages gained have been dramatic." One of the biggest challenges Kumu faced on its legacy provider, for example, was when a celebrity decided to go live on the platform. The app regularly crashed because it was not able to scale fast enough to accommodate the crush of followers.

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Kubernetes

Engine, we're no longer afraid to get celebrities on board. They can stream whenever they want without informing us in advance, and we have no concerns about hitting limitations. That has lowered the barrier of entry for famous personalities to join our platform."

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Kumu tried to get around the problem by requiring famous personalities to inform the platform team in advance before going live, so it could provision additional computing resources. With its original provider, Kumu needed 15 minutes to scale up to

absorb the demand of celebrity fans. Now with GKE, Kumu automatically scales up with no platform outages, then just as easily scales down.

"With Google Kubernetes Engine, we're no longer afraid to get celebrities on board," he says. "They can stream whenever they want without informing us in advance, and we have no concerns about hitting limitations. That has lowered the barrier of entry for famous personalities to join our platform."

For Kumu, one of the most powerful symbols of its success with Google Cloud is the "challenge board" it hangs in its office to mark the number of days in a month the app experienced zero downtime. "In the past, we were never able to fill up the board as we experienced some downtime at least once a week," says Capiral.

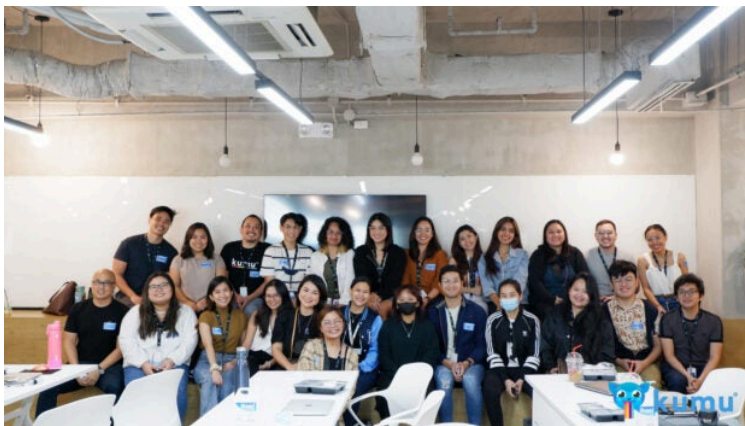
With GKE, the Kumu team has kept the board full week-in week-out, with zero outage days, says Capiral, all while experiencing savings in computing costs and DevOps time. That is thanks to the fact that Kumu no longer needs to spin up computing resources in preparation for a spike, or pay for idling servers waiting for a celebrity to go live. "We only pay for what we use," says Capiral.

## **Delighting audiences with super low-latency live stream delivery**

A major factor in enabling superior user experience with crystal clear live streams, says Capiral, is the switch to Cloud CDN from its legacy content delivery solution. From the beginning, Kumu had set itself a

latency target of a maximum 250 milliseconds. Anything beyond that would cause delays of up to 10 seconds for audience reactions, such as comments or gifts, to appear in the feed.

Since Kumu's popularity depends on user interactions being as real-time as possible, it was a major pain point to see latency regularly go beyond the 250 millisecond limit. This frustrated or even lost viewers in the process.



*Kumu's team*

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"With Cloud CDN, live streaming latency has been reduced to under 100 milliseconds, far under our 250 millisecond target," says Capiral. "This is a huge boost to app performance that translates directly into better audience retention."

With such tangible benefits already being felt in the early stages of a multi-part Google Cloud migration, Kumu is already looking toward future imaginative deployment of Google Cloud solutions to enrich the experience of its global live streaming community.

"We've already seen exciting gains in terms of user engagement, creative capabilities, and platform expansion even in the early days of our Google Cloud journey," says Capiral. "So we're excited to see how we can excite our fans even more through Google Cloud AI transformation."